

Modeling Global Climate Change News: A Computational Analysis of Media Tone Across Countries

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Introduction

As climate crises continue to intensify, they have become a global issue shaped not only by environmental realities but also by how they are represented in the media. News coverage plays a critical role in shaping public understanding by highlighting certain aspects of climate change while downplaying others. These differences in reporting can influence how audiences perceive urgency, responsibility, and potential solutions, ultimately affecting public opinion and policy responses.

Importantly, the framing of climate issues is not uniform across countries. Regional, political, and cultural contexts influence which themes are emphasized, resulting in varied narratives around climate change worldwide. Some countries may focus on economic impacts, while others highlight environmental risks or policy debates. Understanding these differences is essential for identifying how global climate discourse is constructed and communicated.

This project uses global news data to analyze how climate-related themes differ by region, revealing patterns in how climate issues are framed across countries. By examining large-scale news data, this study provides insight into the ways media representation contributes to global climate narratives and highlights the uneven landscape of climate communication worldwide.

Materials / Methods

Data Collection

- Global newsdata obtained from the Global Database of Events, Language, and Tone (GDELT)
- Data accessed using Google BigQuerywith a structured query for climate-related articles

Data Processing

- Analysis conducted in Python using pandas and scikit-learn
- Climate themes extracted from the V2Themes field
- Semicolon-separated themes split into individual entries

Feature Engineering

- Theme frequencies aggregated at the country level
- Data transformed into a wide-format feature matrix
- Rows represent countries and columns represent theme frequencies
- Average tone calculated for each country

Feature Selection

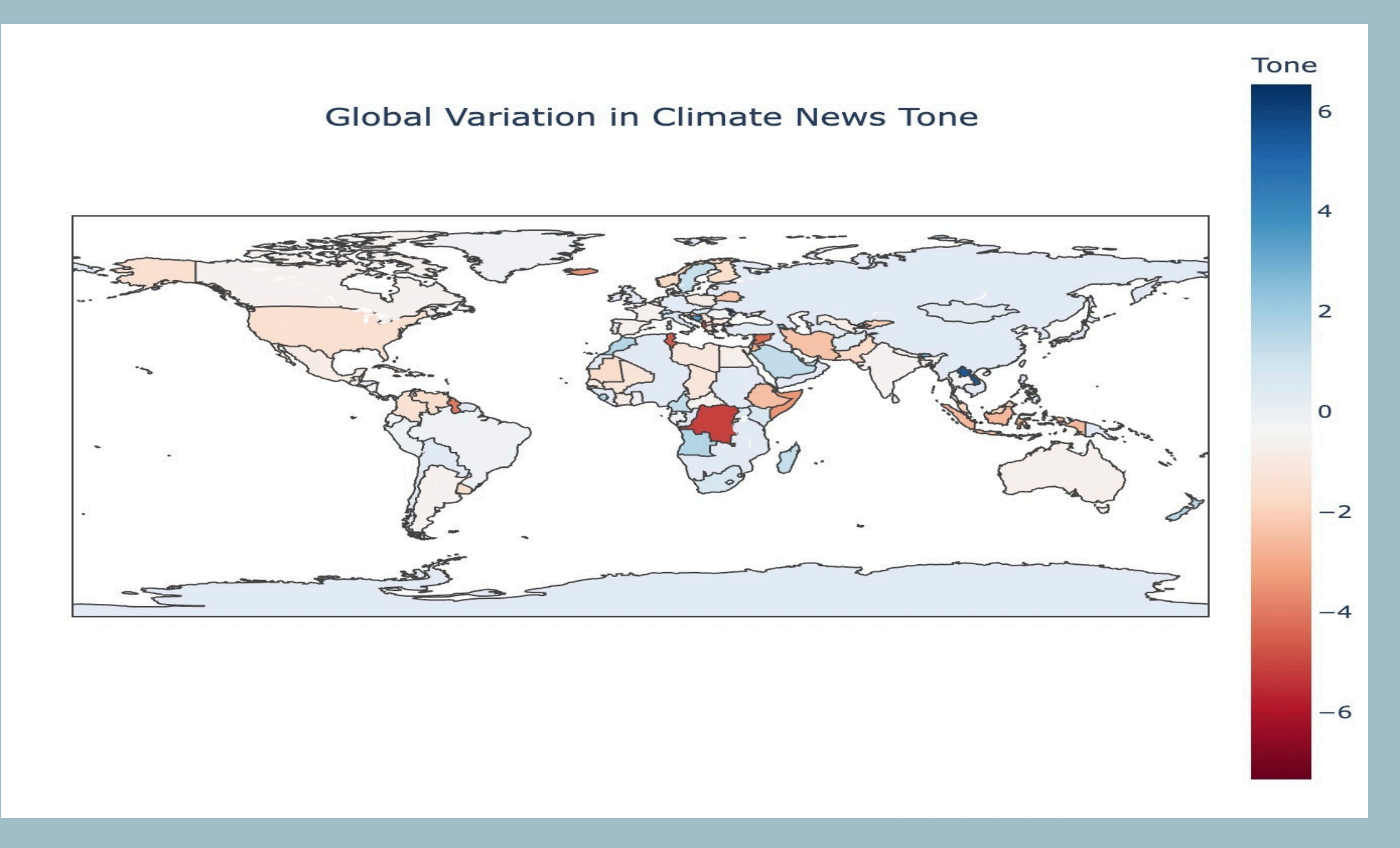
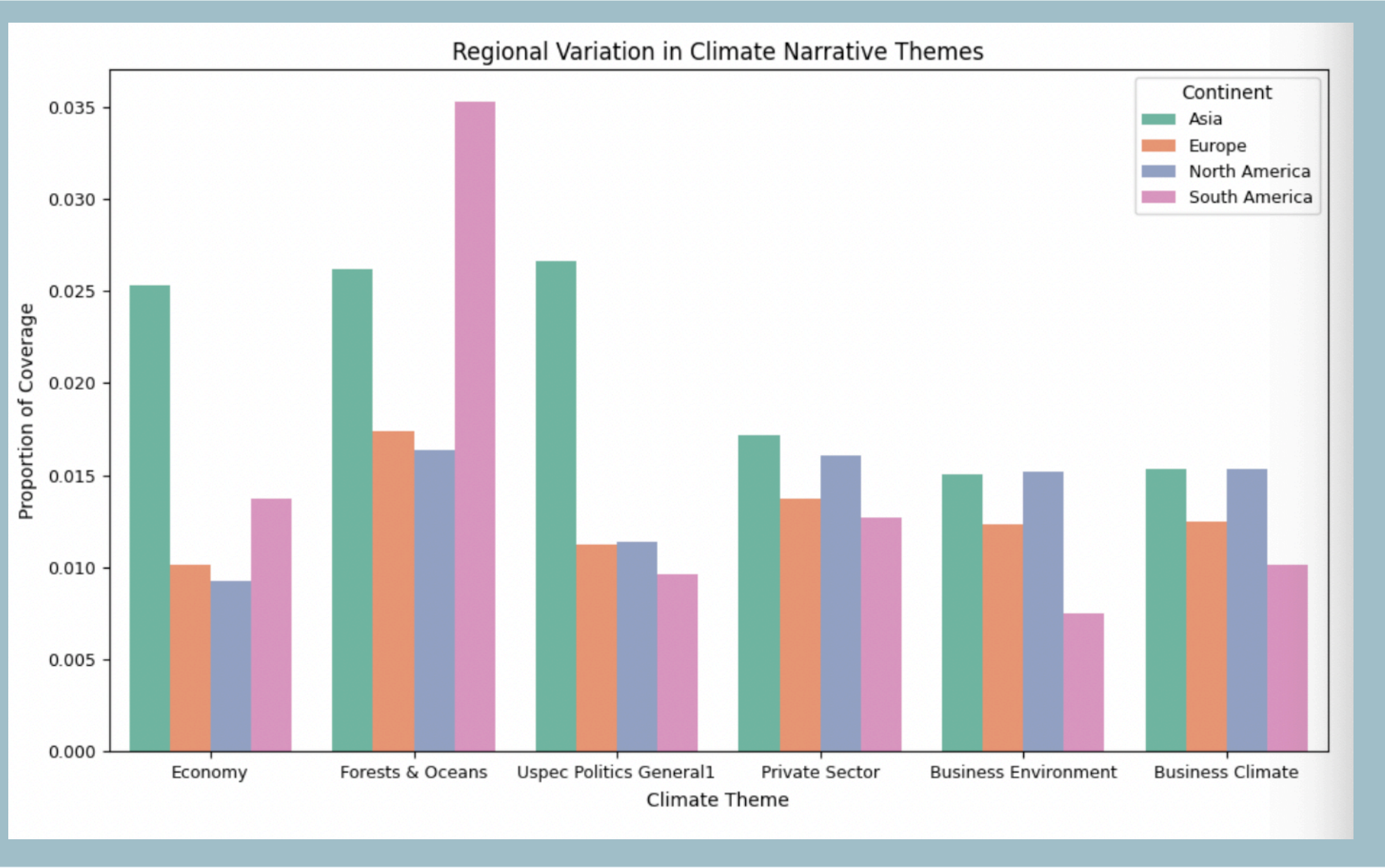
- Initial dataset contained approximately 4,000 features
- Dimensionality reduced through:
 - Variance filtering
 - Sparsity filtering
 - Correlation filtering
- Final feature set reduced to 98 variables

Modeling

- Multiple linear regression used to model tone
- Dataset split into 80% training and 20% testing sets
- Features standardized using StandardScaler

Evaluation

- Model performance assessed using R²



Results

Model Performance

- Final linear regression model included 118 countries and 98 selected features
- Model achieved an R²= 0.365
- Indicates that thematic features explain a moderate portion of variation in country-level tone

Predicted vs. Actual Tone

- Predicted tone values range from -6 to 8
- Actual tone values range from approximately -5 to 3
- Data points follow an overall positive trend
- Some dispersion is present, especially at more extreme values
- A small number of countries show higher predicted tone than observed

Global Tone Distribution (Map).

- Tone varies significantly across countries and regions
- Darker red tones represent more negative values
- Darker blue tones represent more positive values
- Clear regional variation observed across:
 - Africa
 - Asia
 - Europe

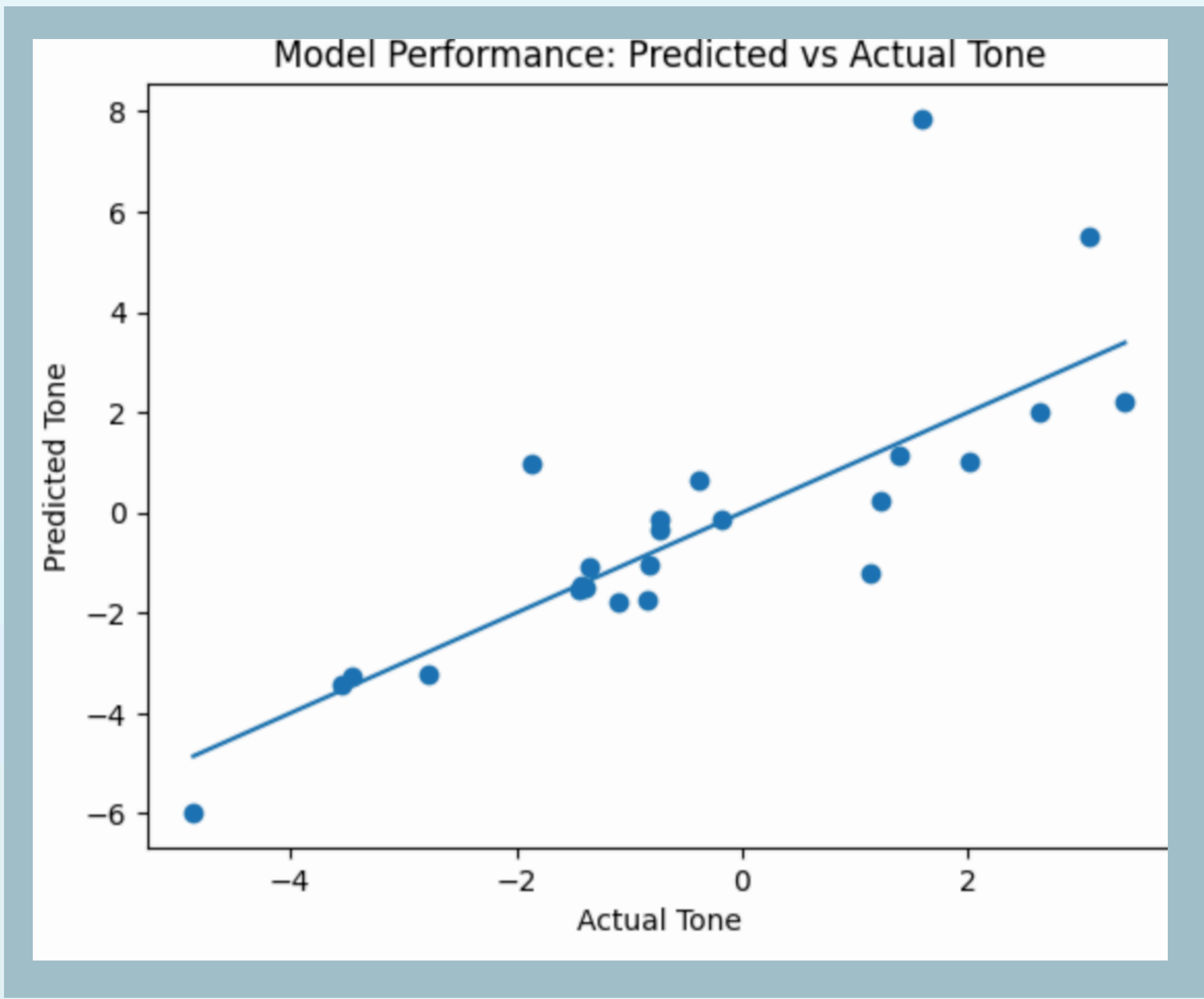
Discussion

This study examines how climate-related themes in global news coverage relate to differences in tone across countries. The results demonstrate that climate narratives vary significantly across geographic regions and are shaped by thematic emphasis in reporting. The global tone map reveals substantial variation across countries, indicating that climate change is framed differently depending on regional context. Parts of Central Africa exhibit more negative tone values, while regions in Europe and Asia tend to show more neutral to slightly positive tones. The United States and several countries in South America appear closer to neutral, highlighting that tone is not consistent even among highly covered regions.

The regression analysis further supports this variation, showing that thematic features are statistically associated with tone. The predicted versus actual tone plot demonstrates a positive relationship, with points generally following an upward trend despite some dispersion. The model achieved an R²value of 0.365, indicating that thematic differences explain a meaningful portion of variation in country-level tone.

The theme-level analysis provides additional insight into how climate narratives differ across regions. The distribution of themes varies across continents, with economic and policy-related topics appearing consistently across regions, while environmental themes such as forests and oceans show greater prominence in certain areas, particularly in South America. These differences indicate that climate coverage is not only varied in tone, but also in the specific topics emphasized, reflecting regional priorities and perspectives in climate discourse.

Taken together, these findings highlight the importance of media framing in shaping global climate narratives. The results indicate that differences in climate narratives are driven not only by geographic variation, but also by the specific themes emphasized in news coverage.



References

AurélienGéron. (2022). *Hands-OnMachine Learning with Scikit-Learn, Keras, and TensorFlow* . O'Reilly Media.

GDELT Project. (n.d.). *GlobalDatabase of Events, Language, and Tone*. Retrieved from <https://www.gdeltproject.org/>

Google BigQuery. (n.d.). *BigQuerypublic datasets*Retrieved from <https://cloud.google.com/bigquery>